

Integrating prediction in mean-variance portfolio optimization

Nonlinear Prediction Models in Integrated Mean-Variance Portfolio Optimization

I. Introduction and Motivation

Portfolio construction traditionally follows a **predict-then-optimize paradigm** in which expected asset returns are first forecast using statistical or machine learning models and subsequently used as inputs to a mean-variance optimization (MVO) problem. While this modular pipeline is intuitive and widely adopted, it implicitly assumes that prediction accuracy translates directly into improved portfolio decisions. However, recent work in decision-focused learning demonstrates that **minimizing prediction error does not necessarily minimize downstream decision cost**.

Integrated Prediction and Optimization (IPO) provides an alternative framework that jointly learns predictive models and decision policies. In the context of portfolio management, **IPO trains return prediction models to directly minimize realized portfolio cost rather than prediction loss**. Prior research has primarily focused on **linear regression models** within the IPO framework due to their analytical tractability and stability in low signal-to-noise financial environments. Nevertheless, modern financial forecasting increasingly relies on **nonlinear prediction architectures** such as neural networks, kernel methods, and tree-based models.

This project investigates whether nonlinear prediction capacity improves decision quality within integrated portfolio optimization. Specifically, we examine whether increased representational flexibility leads to improved out-of-sample portfolio performance when prediction models are trained using decision-focused objectives.

II. Research Question

The central research question of this project is:

Do nonlinear prediction models improve decision quality in integrated portfolio optimization compared to linear models?

To address this question, we consider the following sub-questions:

- Does nonlinear IPO reduce out-of-sample mean-variance portfolio cost?
- How does prediction accuracy relate to downstream portfolio performance?
- Under what data-generating conditions do nonlinear IPO models provide benefits or drawbacks?

III. Background

Mean-Variance Portfolio Optimization

We consider the classical mean-variance objective

$$c(z, y) = -z^\top y + \frac{\delta}{2} z^\top V z, \quad (1)$$

where z denotes portfolio weights, y represents asset returns, V is the covariance matrix, and δ is a risk-aversion parameter.

Integrated Prediction and Optimization

Let $\hat{y} = f(x; \theta)$ denote a predictive model parameterized by θ . The IPO framework solves the bilevel problem

$$\begin{aligned} \min_{\theta} \quad & \mathbb{E}[c(z^*(\hat{y}), y)] \\ \text{s.t.} \quad & z^*(\hat{y}) = \arg \min_z c(z, \hat{y}) \end{aligned} \quad (2)$$

Unlike traditional approaches, the predictive model is trained with respect to realized portfolio cost rather than prediction loss.

Give expected return $\hat{y}$, use it to do mean-variance optimization, get optimized portfolio z^* . Final investment decision resulted in the lowest cost.

IV. Methodology

Baseline Linear IPO

We first implement a baseline linear IPO model where expected returns are predicted via linear regression and trained using the IPO objective. This model serves as the primary benchmark throughout the study.

Nonlinear IPO Extensions

We extend the IPO framework to incorporate nonlinear predictive models, including:

- Feedforward neural networks
- Kernel regression models
- Tree-based ensemble methods

These models will be trained using decision-focused loss derived from realized portfolio performance.

The proposed extension replaces the linear predictive model in the IPO framework with a nonlinear representation while preserving the decision-focused training objective.

$$\text{Basis paper optimization formulation} \quad (3)$$

$$\begin{aligned} \min_{\theta} \quad & \mathbb{E}[c(z^*(\hat{y}), y)] \\ \text{s.t.} \quad & z^*(\hat{y}) = \arg \min_{z \in S} c(z, \hat{y}) \end{aligned} \quad (4)$$

$$c(z, y) = -z^\top y + \frac{\delta}{2} z^\top V z \quad (5)$$

$$\hat{y} = P \text{diag}(x) \theta \quad (6)$$

which corresponds to a linear regression model with asset-specific feature selection.

Here, x denotes the feature vector, θ the regression coefficients, $\text{diag}(x)$ the diagonal feature matrix implementing element-wise scaling, and P a binary matrix selecting asset-specific features. The resulting vector $\hat{y}$ represents predicted asset returns.

Proposed nonlinear IPO formulation (7)

$$\begin{aligned} \min_{\theta} \quad & \mathbb{E}[c(z^*(f(x; \theta)), y)] \\ \text{s.t.} \quad & z^*(f(x; \theta)) = \arg \min_{z \in S} c(z, f(x; \theta)) \end{aligned} \quad (8)$$

Some decisions for $f(x; \theta)$:

- Linear baseline: $f(x; \theta) = Wx + b$
- Polynomial feature model: $f(x; \theta) = W \phi(x) + b$, where $\phi(x) = [x, x^{(2)}, \{x_i x_j\}_{i < j}]$. **【where $\phi(x)$ denotes a nonlinear feature expansion including polynomial and interaction terms.】**
- Kernel predictor: $f(x) = \sum_{i=1}^m \alpha_i K(x, x_i)$
- Neural network predictor : $f(x; \theta) = W_2 \sigma(W_1 x + b_1) + b_2$
- Neural network predictor : $f(x; \theta) = W_3 \sigma(W_2 \sigma(W_1 x + b_1) + b_2) + b_3$
- Asset-wise nonlinear model: $\hat{y}_j = f_j(x_j; \theta_j)$, $j = 1, \dots, d_y$, or $\hat{y}_j = w_j^\top g(x_j; \theta)$.
- Output stabilization: $\hat{y} = \kappa \tanh(g(x; \theta))$

Related works:

- Gu, Kelly, Xiu (2020) — Deep learning in asset pricing
- Krauss et al. (2017) — Deep neural networks for stock prediction
- Fischer & Krauss (2018) — LSTM stock prediction

V. Experimental Design

Data Generating Environments

Experiments will be conducted in two complementary settings:

- Synthetic environment
Synthetic data will be generated with controllable signal-to-noise ratios, nonlinear return structures, and varying asset correlations. This setting enables systematic evaluation of when nonlinear IPO models provide benefits.
- Financial data environment
If feasible, experiments will also be conducted on historical asset data (e.g., equity factors or futures returns) to assess real-world applicability.

Model Comparison

The following models will be compared:

- Predict-then-optimize baselines
- Linear IPO (baseline)
- Nonlinear IPO models

Evaluation Metrics

Performance will be evaluated using:

- Realized mean-variance portfolio cost
- Sharpe ratio
- Prediction accuracy metrics (e.g., MSE, R^2)

Hypothesis Testing

We will analyze whether nonlinear models produce statistically significant improvements in decision metrics and whether gains (if any) correspond to changes in prediction accuracy.

- Expected Contributions

This project is expected to contribute:

- An empirical evaluation of nonlinear prediction models within IPO
- Insights into the relationship between prediction complexity and decision quality
- Practical guidance on model selection for decision-focused portfolio learning